

24SU-EGR150-T101 Final Project: Security Delivery

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1. Scope

This procedure will demonstrate how to use a Sparkfun Micro-Controller to assist with the assembly of the "Secure Delivery" prototype, through the use of a distance sensor, servo motor, temperature sensor, switch, potentiometer, LCD display, LED, along with visual cues. This procedure presupposes a basic understanding of Arduino Coding, and Sparkfun assembly.

2. Reference Documents

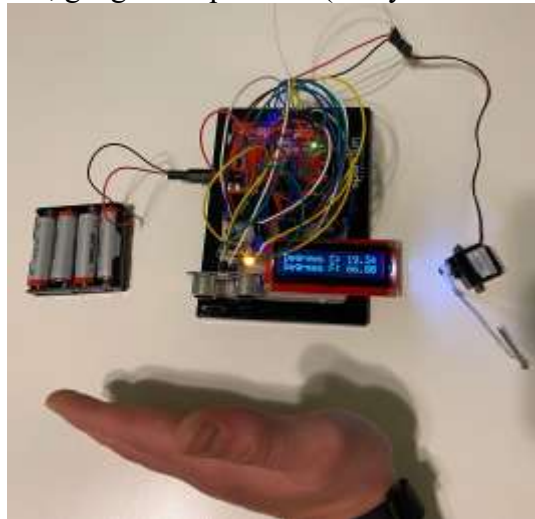
- a. Sparkfun web-page for Kit, as well as tutorials.
 - i. <https://www.sparkfun.com/>
- b. Arduino Software.
 - i. <https://www.arduino.cc/>

3. Terminology

- a. Arduino – Coding language used by RedBoard.
- b. RedBoard1 – Micro-Controller from Sparkfun.
- c. BreadBoard – Circuit building platform.
- d. Servo Motor – 3-Wire motor (power, ground, and signal) that spins 0-180 degrees.
- e. Potentiometer – 3-pin variable resistor.

4. Apparatus

- a. RedBoard Assembly – Determines and gives feedback on distance between delivery driver/homeowner, gauges temperature (every 0.8 seconds) (see Figure 1.)



(Figure 1 - Sparkfun RedBoard)

- b. Computer - used to write the necessary program and upload it to the Sparkfun RedBoard1.
- c. Arduino – Coding language used by RedBoard.
- d. SparkFun Inventor's Kit with components listed.
 - i. Potentiometer
 - ii. Jumper Wires
 - iii. LCD Display
 - iv. Ultrasonic Motion Sensor
 - v. LED
 - vi. Temperature Sensor
 - vii. Battery Converter
 - viii. Switch
 - ix. Servo Motor
 - x. 330 Resistor
- e. Recyclable Materials Used
 - i. Cardboard boxes
 - ii. Packaging tape
 - iii. Paperclip
 - iv. Binder Clip
- f. Miscellaneous (optional)
 - i. Nose needle pliers

5. Procedure

- a. Assemble the Baseplate.
 - i. Mount the breadboard on the Baseplate using the sticky backing.
 - ii. Mount the RedBoard using the provided screws.
- b. Install Arduino Software.
 - i. Visit arduino.cc/downloads.
 - ii. Download and install the Arduino IDE suitable for your computer.
 - iii. Install the necessary drivers from www.sparkfun.com/FTDI.
 - iv. Connect your RedBoard to your computer.
- c. Hookup Assembly
 - i. Jumper Wires: (5V to 5V) (GND to GND(-)) (D8 to E28) (D9 to E27) (D10 to E26) (D11 to E25) (D12 to E20) (D13 to E18) (A0 to E2) (J30 to GND(-)) (J29 to 5V(+)) (J19 to GND(-)) (J16 to 5V(+)) (J15 to GND(-)) (J9 to E17) (J8 to GND(-)) (J10 to 5V(+)) (J1 to GND(-)) (J3 to 5V(+)) (E3 to 5V (+) E6 to GND(-) E24 to GND(-) (I26 to GND (-)) (D13 to J2) (GND to E1)
 - ii. LCD: A15-A30 (pin 1 on A15)
 - iii. LED: A1(-) to A2(+)
 - iv. 330 Resistor: E2 to F2
 - v. Battery Converter: Into designates slot on RedBoard.
 - vi. Temperature Sensor: B1 (GND) + B2 (SIG) + B3(5V
 - vii. Servo Motor: WHITE WIRE to D9 RED WIRE to 5V(+) BLACK WIRE to GND(-)
 - viii. Switch: F25 + F26 + F27

- ix. Ultrasonic Motion Sensor: A3(VCC)+ A4(TRIG)+ A5(ECHO)+ A6(GND)
- x. Potentiometer: A8 + A9 + A10
- c. Write your program.
 - i. Write a suitable program for your project. Examples like the “Distance Sensor”, “Blinking an LED”, "Motion Alarm", "Temperature Sensor", "Motor Basics" will be useful in your writing.
 - ci. Upload your Program to the RedBoard.
 - i. In the Arduino IDE, click Verify to verify your code.
 - ii. Click Upload to send your code to the RedBoard.
 - iii. Watch for the blinking blue light that indicates the RedBoard is receiving data.
 - iv. When done uploading, disconnect your Board from the Computer.
 - cii. Assemble the necessary components.
 - ciii. The circuit used is a combination and remix of Circuits 1A, 3A, 3C, 4B, 5A, as found in the SIK book.

6. Installation

- a. Find cardboard box, paperclip, scissors, Velcro tape, binder clip, needle nose pliers (optional), tape (of any kind).
- b. Use the cardboard to build a box. Within that box cut a hole in the center, a few inches above the bottom of the box. Add in extra spacing using cardboard to elevate the platform that your RedBoard will rest on. The hole is where the motion sensor will peek through. Now make sure that layer is secure using your tape. Then we will grab more cardboard (a smaller box is helpful) using that we will create a protective covering for when the RedBoard is inside the box, cut a section out of the box so you can see the temperature readings. Then find another cardboard piece that fits around the parameter of the inside of the bigger box, make sure to cut a slit in it so you can get light inside the box from the LED, and so you can see the inside temperature. Then tape that piece onto the smaller box that will act as the protection for the RedBoard. Once that is in place, cut out an access panel out of the back of the box for you to easily be able to adjust and take your RedBoard in and out of. Then you will grab your Servo Motor, putting the provided Velcro tape, place it on the bottom of the Servo, then place it on the desired location near the top of your lid. Then grab your paper clip, bend it in a way that it goes around the servo holes. Then grab the binder clip and take apart the metal piece, using that to attach the two to the paperclip so it is one single unit (this is where the needle nose pliers come in hand). Next you want to figure out where it is that your servo needs to be in order for your lid to open and close accordingly (this is a tricky part, use a lighter cardboard piece for the lid if your current lid is too heavy). Once everything is in place with the servo motor and your security box is built out, then we can connect the servo wires to the designated spots, confirm the lid works the way you want, then slide the RedBoard into the access panel with the battery converter. Line up the motion sensor, and turn on your switch. There you go now you have a fully functioning

“security box”. Be sure to make the fitting snug as to ensure the RedBoard doesn't move.

7. Results/Conclusions

Using a Sparkfun RedBoard, a secure delivery box was created to keep online orders of packages/groceries to be hidden, secure, and temperature gaged. With a lid that opens and closes for delivery drivers and homeowners, with an LED light on the inside for when it is dark, and a thermometer to ensure groceries are at a safe temperature displayed on the LCD. One level of distance was utilized to ensure the lid would only open when a person is standing in front of it, and close when the person leaves. The servo motor acted as a lever to open and close the lid. (See Figure 2).



(Figure 2, Servo Motor, Paperclip, Velcro Tape, Binder Clip)

A yellow LED lights up the inside of the box (only when prompted to open), and the Temperature is being constantly checked every 0.8seconds. (See Figure 3).



(Figure 3, LED, and LCD display, looking down into box)

Differing temperatures based on weather outside vs. temperature inside the box at the given time. (Table 1)

Time Temperature was Recorded	00:00 Minutes	5:00 Minutes	10:00 Minutes	20 Minutes
Temperature Inside Box	71.48°F 21.93°C	83.71°F 28.72°C	88.23°F 31.23°C	90.87°F 32.70°C
Temperature Outside	92°F	91°F	91°F	90°F

(Table 1, if a refrigerant system was used for the prototype the box would have stayed much colder during the heat of summer.)

Front view of the security box, Redboard placed so the Motion Sensor is able to read the distance of an object. (Figure 4)



(Figure 4, front view of security box)

When the delivery driver/homeowner walks up to the security box, the box is prompted to open immediately. (See Figure 5).



(Figure 5, delivery driver delivering a package.)

(Figure 5, delivery driver placing in a package, foot in front of motion sensor)
The inside of the box with a package. (Figure 6)



(Figure 6)

Another angle of the box opening. (Figure 6)



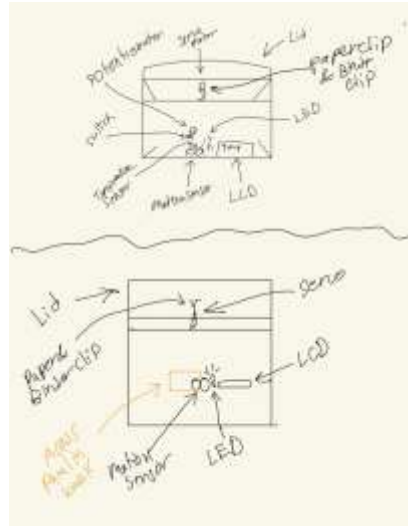
(Figure 6, foot placed in front of motion sensor, delivering package)

Access Panel in back of the security box. (Figure 7)



(Figure 7, access panel)

Sketch of the design. (Figure 8)



(Figure 8, design)

The initial disposition on what the problem was is how often people steal packages right off peoples porches. The next problem was that when ordering groceries online you have to plan your day around when the delivery person arrives so your food isn't sitting outside in the sun all day (doordash is another example). This Project is definitely a "prototype", however, it is a great product which could be an effective way to prevent or cease porch pirates and make online shopping easier. This problem is solved by making a "security box" that is not only discrete but also has a constant eye on the temperature. The results are that the product not only works, but works well. There were a couple of times when the switch was flipped to "on" and it gave an error message over the LCD. However since then, and doing more "test runs" that hasn't happened again. This project was very challenging with all the different codes required, and also trying to fit everything together on the RedBoard/Breadboard. It is a great way to learn more about many different components regarding the Sparkfun kit as well. Starting out one of the hardest things about this project was trying to get the lid to open and close using only the paperclip and binder clip, because it had to be something small enough to go around the servo motor, and string enough to open the lid while also being light enough for it to work. Where it ended was perfect results, while the original plan was to add on a buzzer, that quickly changed when one of the pins/metal wire broke off of it, however, the code for the buzzer is still on the RedBoard just in case. Biggest issue surprisingly wasn't getting the code to work, it was figuring out how to fit everything on the BreadBoard, and finding something to lift the lid most effectively without issues. If anyone repeats this project my advice would be to get the servo motor to work on a much smaller box, then once you see how it works on that one, move it to the bigger box and decide if it needs a lighter lid. This project took around 10-12 hours total more or less, over the course of a few days. The only cost of this project is the SparkFun Kit \$96, other than that the use of free recycled materials around the house does the trick. This project was so fun, during the process of assembly it was noticed that one would continually "get in the zone" and lose track of time. Good luck to anyone else who may want to replicate this project.